

Retinal Fundus Imaging — Complete Optics Reference

All Equations • All Derivations • All Design Decisions

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0. What Signal Are We Trying to Capture?

Before doing any optics, it is worth being precise about what we are actually measuring. The 'signal' is back-scattered near-infrared (NIR) light from the retinal tissue. Here is the chain:

- We illuminate the retina with 850 nm NIR light from the LED ring, entering through the outer annulus of the dilated pupil.
- NIR photons penetrate the transparent ocular media (aqueous, lens, vitreous — all transmit NIR) and reach the retina.
- The retina absorbs some photons (blood vessels are very absorptive; neural tissue less so) and diffusely scatters the rest back out.
- This scattered light exits through the pupil, passes through the condensing lens, forms an aerial intermediate image, then is re-imaged by the 4F relay onto the IMX708 sensor.
- The sensor records pixel intensities proportional to the scattered NIR power. Dark pixels = high absorption (blood vessels, haemorrhages, pigment). Bright pixels = high scattering (nerve fibre layer, drusen, etc.).

What Clinical Features Are Visible in NIR?

Feature	NIR 850nm Appearance	Clinical Relevance
Optic disc	Bright oval; sharply defined margin	Glaucoma: cup-to-disc ratio
Retinal blood vessels	Dark (haemoglobin absorbs NIR)	Diabetic retinopathy, vascular occlusion
Macula/fovea	Slightly darker centre	AMD screening
Choroid	Visible through RPE at 850nm	Choroidal thickness, nevi
Nerve fibre layer	Bright striations	Glaucoma early detection
Drusen (AMD)	Bright deposits	Early AMD
Haemorrhages	Very dark	Diabetic/hypertensive retinopathy

1. Foundational Equations — Full Derivations

1.1 The Thin Lens Equation

All position and magnification calculations in this system derive from the thin lens equation. Using the real-is-positive sign convention (object and image distances measured from the lens, positive to the right):

$$1/f = 1/d_o + 1/d_i$$

Solving for image distance given object distance and focal length:

$$d_i = (f \times d_o) / (d_o - f)$$

Lateral magnification (ratio of image height h_i to object height h_o):

$$M = h_i / h_o = -d_i / d_o$$

The negative sign indicates that a real image ($d_i > 0$) formed by a converging lens with a real object is inverted. If $|M| < 1$ the image is smaller; if $|M| > 1$ it is larger.

1.2 Why the Standard Thin Lens Equation Breaks Down for the Eye

The thin lens equation assumes object and image are both in the same medium (usually air, $n = 1$). The human eye violates this: the retina sits inside vitreous humour at refractive index $n = 1.336$. The correct refraction formula for a single spherical surface is:

$$n_1/s + n_2/s' = (n_2 - n_1) / R$$

where n_1 and n_2 are the refractive indices on either side, s and s' are object and image distances, and R is the radius of curvature. For the full eye we use the Gullstrand simplified model, but there is a cleaner shortcut for our system:

For an emmetropic (perfectly focused) eye, parallel rays from a distant point focus exactly on the retina. By the reversibility of light, any point on the retina emits light that exits the eye as a perfectly parallel (collimated) bundle of rays. This means:

KEY SIMPLIFICATION: The emmetropic eye acts as a collimator. Its output is treated as parallel rays regardless of the internal geometry. The aerial image therefore always forms at exactly f_{cond} behind the condensing lens.

1.3 Full Eye Optical Parameters (Gullstrand Simplified Model)

Parameter	Symbol	Value	Derivation
Total refractive power	P_{eye}	78.6 D	$P = 1000 \times n_{\text{vit}} / f_{\text{post}} = 1000 \times 1.336 / 17.0$
Posterior focal length (in vitreous)	f_{post}	17.0 mm	Retina sits exactly here for emmetropic eye
Anterior focal length (in air)	f_{ant}	12.72 mm	$f_{\text{ant}} = f_{\text{post}} / n_{\text{vit}} = 17.0 / 1.336$
Vitreous refractive index	n_{vit}	1.336	Water-like; roughly constant 380–1400 nm
Axial length (cornea to retina)	L	24.0 mm	Population average; myopes >24mm, hyperopes <24mm
Pupil diameter (natural, dim room)	—	6–8 mm	Undilated: 3–4mm; dilated (pharmacologic): up to 8mm
Cornea to iris (pupil plane)	—	~3 mm	Relevant for illumination angle calculation

1.4 Magnification Formula — Stage 1 (Eye + Condensing Lens)

With the collimated-output approximation, the condensing lens forms the aerial image of a retinal point at its back focal length. The system magnification for Stage 1 is derived from the paraxial ray-trace through the combined eye+condensing system:

$$M_1 = -f_{\text{ant}} / f_{\text{cond}}$$

Substituting values:

$$M_1 = -12.7246 / 35.7143 = -0.3563 \times$$

The minus sign means the aerial image is inverted relative to the retina. The magnitude 0.3563 means the aerial image is 2.81× smaller than the retina.

1.5 Magnification Formula — Stage 2 (4F Relay)

A 4F relay consists of two lenses L1 and L2 with focal lengths f_1 and f_2 . The object (aerial image) is placed at the front focal plane of L1. The magnification is:

$$M_{4F} = -f_2 / f_1$$

Derivation: L1 with object at exactly $d_o = f_1$ gives:

$$1/d_i = 1/f_1 - 1/f_1 = 0 \rightarrow d_i = \infty \text{ (collimated output)}$$

Then L2 with collimated input ($d_o = \infty$):

$$1/d_i = 1/f_2 - 0 = 1/f_2 \rightarrow d_i = f_2$$

So the final image is at f_2 behind L2. The magnification is:

$$M_{4F} = -d_{i_L1} / d_{o_L1} \times (-f_2 / \infty \rightarrow f_1) = -f_2 / f_1 = -100 / 75 = -1.333 \times$$

1.6 Total System Magnification

$$M_{\text{total}} = M_1 \times M_{4F} = (-0.3563) \times (-1.333) = +0.4751 \times$$

Both stages invert; the double inversion means the final sensor image is upright relative to the retina. The total magnification is 0.4751×, meaning the sensor image is about half the size of the retina.

1.7 Radial Offset Derivation — LED Ring Inner Diameter

The goal is to illuminate the retina through the outer annulus of the pupil, while the camera collects through the inner zone. We need the inner radius of the LED ring to be larger than the camera's collection beam at the condensing lens plane.

Step 1 — Find the camera's collection numerical aperture at the cornea:

$$NA_{\text{eye}} = r_{\text{pupil}} / L_{\text{axial}} = 4.0 \text{ mm} / 24.0 \text{ mm} = 0.1667$$

Step 2 — Project this NA through the condensing lens to find the beam radius at the condensing lens plane:

$$\begin{aligned} r_{\text{beam}} &= NA_{\text{eye}} \times f_{\text{cond}} = 0.1667 \times 35.71 = 5.95 \text{ mm} \\ d_{\text{beam}} &= 2 \times 5.95 = 11.90 \text{ mm} \text{ (imaging beam diameter)} \end{aligned}$$

Step 3 — Set ring inner radius to 1.2× the beam radius (20% margin to prevent vignetting):

$$\begin{aligned} r_{\text{inner}} &= 1.2 \times r_{\text{beam}} = 1.2 \times 5.95 = 7.14 \text{ mm} \rightarrow d_{\text{inner}} \\ &= 14.29 \text{ mm} \end{aligned}$$

Step 4 — Add ring width (3 mm) for outer radius:

$$r_{\text{outer}} = r_{\text{inner}} + 3.0 = 10.14 \text{ mm} \rightarrow d_{\text{outer}} = 20.29 \text{ mm}$$

Step 5 — Verify angles at cornea are within dilated pupil acceptance:

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θ_inner = arctan(r_inner / WD) = arctan(7.14 / 10.0) = 35.5°
θ_outer = arctan(r_outer / WD) = arctan(10.14 / 10.0) = 45.4°
θ_pupil ≈ arctan(r_pupil / d_cornea-iris) = arctan(4 / 3) = 53.1°

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RESULT: Both ring edges (35.5° and 45.4°) are within the 53.1° pupil acceptance angle. All LED light enters the eye. No vignetting.

2. Complete Optical Train — All Positions

Element	z from Cornea	Derivation	Movement
Cornea (eye phantom)	0 mm	Fixed datum	Fixed
28D condensing lens + LED ring	10.0 mm	WD = 10mm (working distance)	Moves together as 1 unit ±5mm
Aerial intermediate image	45.71 mm	WD + f_cond = 10 + 35.71	Virtual (no optic here)
L1 (f ₁ = 75mm achromat)	120.71 mm	z_aerial + f ₁ = 45.71 + 75	Independent ±10mm
Collimated space between L1–L2	120.71 – 295.71 mm	Length = f ₁ + f ₂ = 75 + 100 = 175mm	—
L2 (f ₂ = 100mm achromat)	295.71 mm	z_L1 + f ₁ + f ₂ = 120.71 + 175	Independent ±10mm
IMX708 sensor	395.71 mm	z_L2 + f ₂ = 295.71 + 100	Independent ±5mm (fine focus)

⚠ All 4 elements (condensing unit, L1, L2, camera) **MUST** be individually slideable on the rail. Locking any two together removes a degree of freedom you will need to compensate for lens manufacturing tolerances (±1–2% in focal length).

3. Sensor Calculations — IMX708

Parameter	Value	Derivation
Sensor model	Sony IMX708	RPi Camera Module 3 spec sheet
Array	4608 × 2592 pixels	—
Pixel pitch	1.4 μm	—
Sensor width	6.451 mm	4608 × 0.0014
Sensor height	3.629 mm	2592 × 0.0014
Sensor diagonal	7.40 mm	√(6.451 ² + 3.629 ²)
Macula image on sensor	2.61 mm (72% height)	5.5 mm × M_total = 5.5 × 0.475

Parameter	Value	Derivation
Fovea image on sensor	0.713 mm (20% height)	1.5 mm × 0.475
Optic disc image on sensor	0.855 mm (24% height)	1.8 mm × 0.475
Retinal area per pixel	2.95 μm/px	(1.4 μm / 0.475) × 1
Foveal photoreceptor spacing	~2.5 μm	Literature; below our 2.95 μm/px limit
Macular cone spacing	~5–10 μm	Well above Nyquist limit — resolvable

Resolution interpretation: our system resolves features down to ~3 μm on the retina. Foveal cones (2.5 μm) are just below the Nyquist limit — this is fine because the clinical targets (blood vessels, optic disc margin, drusen) are all 50–200+ μm. We have more than sufficient resolution for the intended diagnostic purpose.

4. The Pocket Retina 28D Lens — Is It Suitable?

4.1 Glass Transmission vs Wavelength

The Pocket Retina 28D is manufactured from optical glass (N-BK7 or equivalent borosilicate). The Sellmeier-derived refractive indices for N-BK7 are:

Wavelength	n (N-BK7)	Transmission	Notes
450 nm (blue visible)	1.5282	>92%	Standard visible range
550 nm (green visible)	1.5187	>92%	Peak photopic sensitivity
650 nm (red visible)	1.5151	>92%	—
850 nm (NIR — our choice)	1.5100	>91%	Slight index shift only
940 nm (deep NIR)	1.5082	>90%	Lower IMX708 sensitivity
1064 nm (deep NIR)	1.5066	>88%	Out of sensor range

4.2 Chromatic Aberration (Focal Shift with Wavelength)

Because n changes with wavelength (dispersion), the focal length $f = R/(n-1)$ for a singlet also changes. The focal shift between two wavelengths is approximately:

$$\Delta f \approx -(\Delta n / n) \times f$$

For our condensing lens ($f = 35.71$ mm) from 550 nm to 850 nm:

$$\Delta n = n(550\text{nm}) - n(850\text{nm}) = 1.5187 - 1.5100 = +0.0087$$

$$\Delta f = -(0.0087 / 1.5144) \times 35.71 = -0.21 \text{ mm}$$

A 0.21 mm shift in aerial image position is well within the depth of focus of the system (several mm), so the lens works correctly at 850 nm without any focus adjustment.

VERDICT: Yes. The Pocket Retina 28D lens works for 850nm, 940nm, and visible light. The focal shift is 0.21mm — negligible. The only practical limitation is the AR coating efficiency (designed for visible; slightly less effective at NIR, causing ~1–4% extra surface reflection). This is acceptable for a prototype.

4.3 850 nm vs 940 nm — Which Should You Use?

Factor	850 nm	940 nm
IMX708 quantum efficiency (estimated)	~30–40%	~10–15%
Pupil constriction (photoreceptor response)	Minimal	None
N-BK7 transmission	>91%	>90%
Atmospheric water absorption band	Not affected	Minor absorption (~5%)
LED availability (SMD 5050)	Abundant	Abundant
LED driver power needed for equal SNR	1×	~3–4×
Recommendation	✓ USE THIS	Fallback only

5. Camera Focus — Do You Need to Unscrew It?

The Shen & Mukai (2017) paper physically unscrewed the NoIR camera module lens counterclockwise using jewellery pliers to change its focus from infinity to ~8 cm. This is risky and was necessary only because of their specific setup. Here is a full comparison:

	Shen & Mukai 2017	Your System
Relay lenses?	None — camera images aerial image directly	Yes — 4F relay (L1 + L2)
What the camera 'sees'	Aerial image ~11cm away	Collimated beam from L2
Default focus (infinity) OK?	No — infinity focus cannot see 11cm object	YES — 4F outputs collimated light
Fix required	Physically unscrew lens to refocus at 8cm	Software: set LensPosition = 0
Risk	HIGH — scratch lens, break flex cable, void warranty	None

VERDICT: Do NOT unscrew the camera. Your 4F relay outputs collimated light directly into the camera, which the camera lens re-focuses onto the IMX708 at infinity. Set manual focus via software only.

Software command:

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rpicam-still --autofocus-mode manual --lens-position 0 -o retina.jpg
# Or with exposure control:
rpicam-still --autofocus-mode manual --lens-position 0 --shutter
5000 --gain 8 -o retina.jpg
```

If images appear blurry at LensPosition=0, try values 0.0 to 0.3 (smaller = farther focus). The 4F geometry is designed for infinity, but manufacturing tolerances in f_1 and f_2 may cause a small offset.

6. Glare Management and Polarization

6.1 The Corneal Reflex Problem

When light hits the cornea, approximately 4% reflects back (Fresnel reflection at air/ $n=1.376$ interface). Without mitigation this creates a brilliant white spot on the sensor that overwhelms the much dimmer retinal signal. All non-mydriatic fundus cameras must address this.

6.2 Annular Illumination (Our Primary Solution)

The illumination beam enters the pupil through the outer annular zone (ring radii 7.14–10.14 mm at the condensing lens plane, corresponding to 35.5–45.4° at the cornea). The camera collects through the inner zone (< 5.95 mm radius at condensing plane, < ~30° at cornea). The corneal reflex from the ring LEDs reflects back outward — spatially separated from the camera's collection cone — and never reaches the sensor.

This is the same principle used in all modern non-mydriatic fundus cameras. It requires no additional components beyond the ring geometry we have already designed.

6.3 Polarization — When and How to Add It

Cross-polarization is a secondary technique that removes specular reflections (cornea, crystalline lens) that survive the spatial separation:

- Place a linear polarizer film over the LED ring (orientation: 0°).
- Place a second linear polarizer over the camera entrance, rotated 90° (crossed).
- Specular reflections (mirror-like bounces off corneal surface) preserve polarisation — blocked by the crossed analyser.
- Retinal scatter depolarises light — partially passes through the analyser (~50% transmitted).

⚠ Standard visible-light polarisers are opaque at 850 nm. You MUST use NIR-rated polarisers (e.g. Thorlabs LPNIRE050-B, or wire-grid polarisers rated 700–1000nm). Using wrong polarisers will block ALL your imaging light.

RECOMMENDATION: Try without polarisers first. Annular illumination should handle the corneal reflex adequately. Add crossed NIR polarisers only if glare artifacts remain after initial testing with the phantom.

6.4 All Glare Sources — Causes and Fixes

Glare Source	Physical Cause	Mitigation
Corneal anterior surface	Fresnel reflection ~4% at air-cornea interface	PRIMARY: annular illumination. SECONDARY: crossed NIR polarisers

Glare Source	Physical Cause	Mitigation
Crystalline lens (Purkinje III & IV)	Reflections from anterior/posterior lens surfaces	Annular illumination + polarisation
Condensing lens surfaces	~1–4% per surface without NIR AR coat (4 surfaces total)	NIR AR-coated lens; black lens barrel; absorbing baffle tube
4F relay internal scatter	Stray light bouncing L1↔L2 in collimated space	AR-coated achromats (B-coat); black tube housing; aperture stop in collimated space
LED ring PCB backscatter	Light reflected off PCB soldermask into L1	Black soldermask; anodised black ring mount; mechanical baffle
Ambient room light	Room light entering from sides of system	Dark enclosure tube from condensing lens to camera; test in dim room
Camera module internal flare	Reflections within RPi camera's own lens barrel	Remove camera lens (advanced) or use lens hood; minimal at small aperture

7. All Equipment Questions Answered

7.1 Do You Need a Macro Lens?

A 'macro lens' is a camera lens designed to focus at very short distances, producing life-size (1:1) or larger magnification. Normal camera lenses cannot focus closer than ~30 cm.

Do you need one? No. Your 4F relay does the close-range imaging. The camera receives collimated light from L2 and uses its own lens set to infinity. The camera lens does not need macro capability.

7.2 Do You Need a Bigger Lens?

The 25.4 mm (1") diameter achromats for L1 and L2 are sized against the imaging beam diameter at those planes. At L1, the beam from the 11.9 mm diameter at the condensing lens is collimated — it stays 11.9 mm in diameter through the entire L1-L2 space. The 25.4 mm aperture is 2.1× larger than the beam: comfortable clearance with no vignetting.

2" (50.8 mm) lenses would give more margin and potentially less edge aberration, but at 2–3× the cost. Not necessary for the prototype.

7.3 Pinhole Camera?

A pinhole camera uses a tiny aperture instead of a lens — infinite depth of field but extremely low light throughput. Not appropriate here. Your lens system collects vastly more light and produces a proper focused image. A pinhole would require damagingly bright illumination.

7.4 Pinhole Tube / Dark Tube Between Ring and Eye?

This refers to a cylindrical shroud or baffle around the front of the instrument (condensing lens + LED ring). It is NOT for imaging — it is for ambient light exclusion and glare reduction. Clinical fundus cameras all have a 'nose piece' or snout of this type.

For your prototype: a 3D-printed matte-black tube that slips over the condensing lens assembly and extends ~5 mm toward the eye is cheap and effective. Paint or print the inside in matte black (PLA naturally rough surfaces work). This is highly recommended — it will meaningfully improve image contrast.

7.5 Beam Splitter?

A beam splitter splits the illumination and imaging paths coaxially (same axis). Used in direct ophthalmoscopes and scanning laser ophthalmoscopes. It costs 50% of the illumination AND 50% of the return signal = 75% total light loss. More complex, more reflections.

VERDICT: Not needed. Annular illumination spatially separates the paths with zero light loss penalty.

7.6 IR Lamp?

A broad-spectrum infrared lamp would provide wide-band NIR illumination. Not suitable — it wastes power, reduces contrast (broad spectrum reduces signal-to-background), and the IMX708 has low quantum efficiency at longer IR wavelengths. A narrowband 850 nm LED ring provides targeted, efficient illumination exactly where the sensor is most sensitive.

7.7 Singlet vs Achromatic Doublet for the 4F Relay?

Spherical aberration for a singlet plano-convex lens scales as $D^4/(128 \cdot n \cdot f^3)$. For our 25.4mm diameter, 75mm focal length L1:

$$\Delta f_{\text{spherical}} = D^4 / (128 \times n \times f^3) = 25.4^4 / (128 \times 1.515 \times 75^3) = 0.005 \text{ mm}$$

This is tiny — spherical aberration is not a problem at 1" diameter. Chromatic aberration for a singlet: our illumination is narrowband 850 nm (LED bandwidth ~20–30 nm), so chromatic shift is < 0.01 mm. Singlets would likely work.

However achromatic doublets are the professional choice, correct aberrations over a range of wavelengths, and cost ~\$60–85 each from Thorlabs. If budget allows, use achromats. Specify B-coat (Thorlabs part suffix -B) which is AR-optimised for 650–1050 nm.

8. Myopia, Hyperopia, and Refractive Error Compensation

An emmetropic (normal) eye has axial length 24 mm — the retina sits exactly at the focal plane. Myopic eyes are longer (e.g. 26 mm); hyperopic eyes are shorter (e.g. 22 mm). In both cases the retina is NOT at the back focal plane, so light does NOT exit as parallel rays.

Effect on our system: the aerial image shifts axially (closer or farther than f_{cond} behind the condensing lens). The 4F relay then receives the object at the wrong position, producing a blurred or out-of-focus image on the sensor.

Corrections in order of complexity:

- Manual working distance (WD) adjustment: moving the entire condensing+ring unit closer or farther shifts the aerial image. Adjust until the image on the RPi preview is sharpest. This works for moderate refractive errors ($\pm 5D$).
- Sensor position adjustment: fine-sliding the camera along the rail can compensate for small focal plane shifts.
- Liquid lens (future work): an electrically tunable lens (e.g. Optotune EL-10-30 or Corning Varioptic Arctic series) inserted between the condensing lens and L1 can electrically shift the aerial image position to compensate any refractive error from $-10D$ to $+10D$. Out of scope for BME 462 v1.

For your 60D eye phantom: it models an emmetropic eye — no refractive error compensation needed during initial testing. The retina (printed target) is placed 16.7 mm behind the 60D lens (= $1000/60$ mm = back focal length).

9. What Professional Retinal Cameras Do (That We Are Simplifying)

Feature	Professional Camera	Our Prototype
Patient alignment	Motorized XYZ joystick stage; chin rest; forehead rest	Manual rail adjustment; 3D-printed chin rest (recommended)
Fixation target	Internal LED; patient fixates to centre gaze on fovea	Not implemented; test with eye phantom only
Refractive error correction	Built-in diopter wheel $\pm 12D$	Manual WD adjust; liquid lens as future upgrade
Autofocus	Motorized; contrast-detection or wavefront sensing	Manual L1/L2/camera position on optical rail
Field of view	Selectable: 30° / 45° / 60° / 100°	Fixed ~30–35° with 28D condensing lens
Image capture	Synchronized NIR preview → white flash → colour image	NIR only (greyscale); no colour fundus mode
Light safety certification	EN ISO 15004-2 (retinal hazard group assessment)	Must calculate MPE; informal safety estimate only
Fluorescein angiography	488 nm excitation + 520 nm barrier filter	Not implemented
OCT imaging	Swept-source or spectral-domain OCT, μm axial resolution	Not implemented
Wide-field imaging	Optos: scanning laser, 200° FOV	Not implemented (~30° only)
DICOM / EHR integration	Standard hospital PACS connectivity	Raw JPEG from RPi
Portability	Table-mounted, ~20–40 kg	Handheld benchtop, ~1–2 kg target
Unit cost	\$10,000 – \$50,000 USD	< \$300 CAD target

9.1 Light Safety — Why You Must Calculate It

NIR light is invisible. The eye cannot blink to protect itself (no aversion reflex). Retinal damage from NIR is possible at sustained exposures — it is thermal damage to the RPE and photoreceptors.

The relevant standards are ANSI Z136.1 (laser/LED safety) and EN ISO 15004-2 (ophthalmic instruments). The key quantity is the Maximum Permissible Exposure (MPE) for the retina at 850 nm for your exposure duration.

For a rough estimate: the Shen & Mukai paper measured 50–90 μW at 850 nm at the corneal plane. A Keeler clinical indirect ophthalmoscope produces 1500–3000 μW at 570 nm. Their prototype was ~20–30× below clinical instruments. Your LED ring at typical drive currents will be in a similar range — well below clinical limits for a 1-second imaging session.

⚠ This calculation must appear in your BME 462 Design Report under 'Safety and Regulatory Considerations.' It is not optional for a medical device prototype. Estimate LED power, calculate irradiance at cornea, compare to ANSI/ISO MPE for 850 nm.

10. Complete System Parameter Summary

Parameter	Value	Key Equation / Note
Eye power (Gullstrand model)	78.6 D	$P = 1000 \times 1.336 / 17.0$
Eye anterior focal length (in air)	12.72 mm	$f_{\text{ant}} = f_{\text{post}} / n_{\text{vit}} = 17.0 / 1.336$
Condensing lens power	28 D	Pocket Retina 28D
Condensing lens focal length	35.71 mm	$f = 1000 / 28$
Working distance (WD) from cornea	10 mm	Adjustable $\pm 5\text{mm}$
Aerial image distance from cornea	45.71 mm	$WD + f_{\text{cond}}$
Stage 1 magnification	$-0.356\times$	$-f_{\text{ant}} / f_{\text{cond}}$
L1 focal length	75 mm	Thorlabs AC254-075-B
L2 focal length	100 mm	Thorlabs AC254-100-B
z_L1 from cornea	120.71 mm	$z_{\text{aerial}} + f_1$
L1–L2 separation	175 mm	$f_1 + f_2$
z_L2 from cornea	295.71 mm	$z_{\text{L1}} + f_1 + f_2$
z_sensor from cornea	395.71 mm	$z_{\text{L2}} + f_2 (\approx 40 \text{ cm total})$
4F relay magnification	$-1.333\times$	$-f_2 / f_1$
Total system magnification	$+0.475\times$	$M_1 \times M_4F$ (double inversion $\rightarrow$ upright)
Sensor (IMX708) size	$6.45 \times 3.63 \text{ mm}$	$4608 \times 2592 \text{ px}$, $1.4 \mu\text{m/px}$
Macula image on sensor	2.61 mm (72% height)	$5.5 \text{ mm} \times 0.475$
Retinal resolution per pixel	$2.95 \mu\text{m}$	$1.4 \mu\text{m} / 0.475$
LED ring inner diameter	14.29 mm	$2 \times 1.2 \times \text{NA}_{\text{eye}} \times f_{\text{cond}}$
LED ring outer diameter	20.29 mm	inner + 6mm ring width
Illumination angle at cornea	$35.5^\circ - 45.4^\circ$	Within 53° dilated pupil: ✓
Illumination wavelength	850 nm NIR	Best IMX708 QE; minimal pupil response
Camera focus setting	Manual, LensPosition = 0 (infinity)	Do NOT physically unscrew lens
Camera variant required	RPi Camera Module 3 NoIR	Standard variant IR filter blocks 850 nm
4F lenses	Achromatic doublets, B-coat AR	B-coat = 650–1050 nm optimised
Polarisation	Not required for v1	Add crossed NIR polarisers if glare persists

Parameter	Value	Key Equation / Note
Beam splitter	Not used	Annular illumination replaces it
Eye phantom retina distance	16.7 mm behind 60D lens	$f = 1000/60 = 16.67 \text{ mm}$